Seurat Workflow on PBMC3k: Aim, Workflow and Interpretation of Plots

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1. Aim

To analyse a single-cell RNA-seq dataset of about 3,000 human peripheral blood mononuclear cells (PBMC3k, 10x Genomics, hg19) using the Seurat package in R, and to move step by step from the raw count matrix to biologically labelled cell populations. The specific objectives are:

- To load the raw 10x count matrix into a Seurat object and understand its sparse structure.
- To perform quality control (QC) and remove low-quality cells (empty droplets, doublets, dying cells).
- To normalise the data, identify highly variable genes and scale the data.
- To reduce dimensionality using PCA and decide how many principal components to keep.
- To cluster the cells, visualise them with UMAP, and find the marker genes of every cluster.
- To assign cell-type names to clusters using known marker genes.

2. Overall workflow

Part A (terminal): The file pbmc3k_filtered_gene_bc_matrices.tar.gz was checked, extracted into data/pbmc3k, and the three expected files (barcodes.tsv, genes.tsv, matrix.mtx) were confirmed inside filtered_gene_bc_matrices/hg19/. A results folder was created and R was started from the bio-env directory.

Part B (R, Seurat):

1. Read the data with Read10X and create the Seurat object (CreateSeuratObject, min.cells = 3, min.features = 200).
2. Quality control: calculate percent.mt, view violin and scatter plots, and filter cells (nFeature_RNA between 200 and 2500, percent.mt below 5).
3. Normalisation with NormalizeData (LogNormalize, scale factor 10,000).
4. Selection of 2,000 highly variable genes with FindVariableFeatures (vst method).
5. Scaling of all genes with ScaleData.
6. PCA with RunPCA, followed by VizDimLoadings, DimPlot and DimHeatmap to inspect the PCs.
7. Choice of dimensionality using ElbowPlot.
8. Graph-based clustering with FindNeighbors (dims 1:10) and FindClusters (resolution 0.5).
9. UMAP embedding with RunUMAP (dims 1:10).
10. Cluster biomarkers with FindMarkers and FindAllMarkers, checked with violin plots, feature plots and a heatmap.
11. Renaming clusters with cell-type labels (RenameIdents) and plotting the final labelled UMAP.

3. Section 1: Setting up the Seurat object

```
    intersect, t

v5.6 adds significant improvements in speed and efficiency, particularly for large datasets.
For more information, see https://satijalab.org/seurat/articles/announcements.

Keep up to date with changes at https://tidyverse.org/blog/
Warning: Feature names cannot have underscores ('_'), replacing with dashes ('-')
An object of class Seurat
13714 features across 2700 samples within 1 assay
Active assay: RNA (13714 features, 0 variable features)
 1 layer present: counts
>
```

Output 1: Seurat object after loading the data

The object was created with 13,714 genes (features) across 2,700 cells in one assay (RNA), with only the raw counts layer present. R also printed a warning that gene names cannot contain underscores, so they were replaced with dashes. This is only a naming fix and does not change the data.

```
sparse.size

dense.size / sparse.size
3 x 30 sparse Matrix of class "dgCMatrix"
  [[ suppressing 30 column names ‘AAACATACAACCAC-1’, ‘AAACATTGAGCTAC-1’, ‘AAACATTGATCAGC-1’ ... ]]

CD3D  4 . 10 . . 1 2 3 1 . . 2 7 1 . . 1 3 . 2  3 . . . . . 3 4 1 5
TCL1A . .  . . . . . 1 . . . . . . . . . . . .  . 1 . . . . . . . .
MS4A1 . 6  . . . . . 1 1 1 . . . . . . . . . . 36 1 2 . . 2 . . . .
709591472 bytes
29905192 bytes
23.7 bytes
>
```

Output 2: raw counts of CD3D, TCL1A and MS4A1 in the first 30 cells, and dense versus sparse memory size

Most entries in the count table are dots. A dot is a zero, meaning the gene was not detected in that cell. CD3D (a T cell gene) has non-zero counts in many of the 30 cells, while TCL1A and MS4A1 (B cell genes) are mostly zero but show a few cells with counts, and MS4A1 reaches high values (such as 36) in what are probably B cells. This is typical of single-cell data, where each cell expresses only a fraction of all genes and dropouts are common.

The dense matrix needs 709,591,472 bytes (about 710 MB), whereas the sparse matrix needs only 29,905,192 bytes (about 30 MB). The ratio is 23.7, so the sparse format is about 24 times smaller. Seurat stores only the non-zero values and their positions, and since the great majority of the matrix is zero, a large amount of memory is saved. This is why large single-cell datasets can be analysed on an ordinary computer.

4. Section 2: Quality control and selection of cells

```
Active assay: RNA (13714 features, 0 variable features)
 1 layer present: counts
> # Mitochondrial percentage
pbmc[["percent.mt"]] <- PercentageFeatureSet(pbmc, pattern = "^MT-")

# Show QC metrics for the first 5 cells (was missing)
head(pbmc@meta.data, 5)

# Violin plot of QC metrics
VlnPlot(pbmc, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
                orig.ident nCount_RNA nFeature_RNA percent.mt
AAACATACAACCAC-1     pbmc3k       2419          779  3.0177759
AAACATTGAGCTAC-1     pbmc3k       4903         1352  3.7935958
AAACATTGATCAGC-1     pbmc3k       3147         1129  0.8897363
AAACCGTGCTTCCG-1     pbmc3k       2639          960  1.7430845
AAACCGTGTATGCG-1     pbmc3k        980          521  1.2244898
Warning: Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead.
>
```

Output 3: QC metrics of the first five cells

The metadata table gives, for each cell, nCount_RNA (total RNA molecules), nFeature_RNA (number of different genes detected) and percent.mt (percentage of reads from mitochondrial

genes). The first cell has 2,419 counts, 779 genes and 3.02% mitochondrial reads, and the third cell has 3,147 counts, 1,129 genes and 0.89% mitochondrial reads. All five cells have low mitochondrial percentages (about 0.9 to 3.8%), which indicates healthy, intact cells.

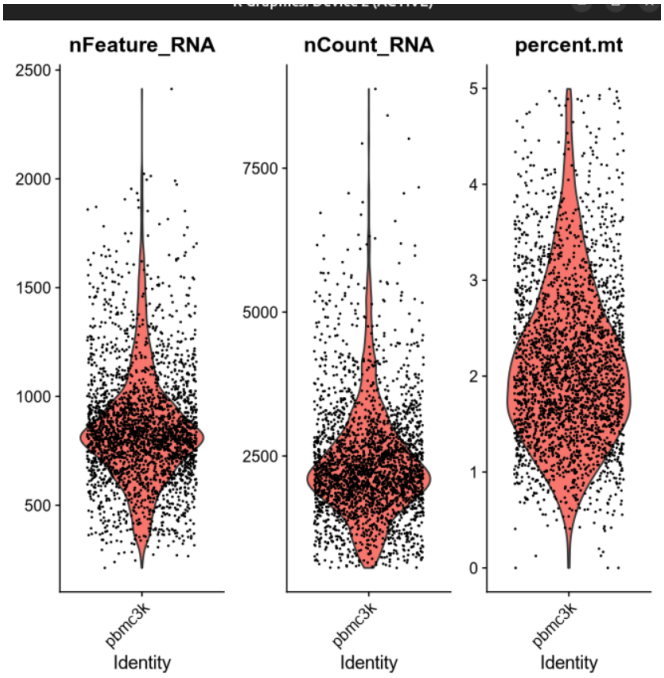


Figure 1: Violin plots of nFeature_RNA, nCount_RNA and percent.mt

- **nFeature_RNA:** most cells have between about 500 and 1,200 detected genes, and the widest part of the violin is near 800. A few cells reach about 2,000 to 2,400 genes. Cells with very few genes are usually empty droplets or damaged cells, and cells with very many genes may be doublets. This is the reason for the lower cut-off (200) and upper cut-off (2,500).
- **nCount_RNA:** most cells have about 1,500 to 3,500 counts, with the widest part near 2,200, and a long tail of a few cells up to about 8,000. The shape follows nFeature_RNA, because cells with more counts also have more genes.
- **percent.mt:** values are spread between 0 and 5%, with most cells between 1 and 3%. A high mitochondrial percentage indicates dying cells whose cytoplasmic RNA has leaked out, so the cut-off of 5% removes such cells.

The upper limits visible on the plot (about 2,500 genes and 5% mitochondrial reads) match the filtering thresholds, so these violins represent the cells retained after filtering.

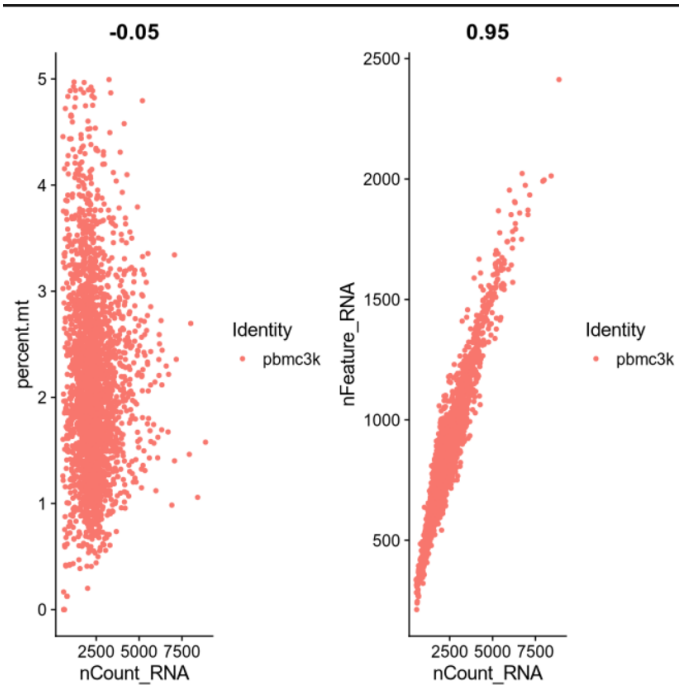


Figure 2: FeatureScatter plots (percent.mt vs nCount_RNA, and nFeature_RNA vs nCount_RNA)

- **percent.mt vs nCount_RNA:** the correlation is -0.05, essentially zero. Mitochondrial percentage does not depend on sequencing depth, so filtering on percent.mt does not bias the data towards low or high count cells.
- **nFeature_RNA vs nCount_RNA:** the correlation is 0.95, a very strong positive relationship. Cells with more total counts have more detected genes, and the points fall along a tight, almost linear band. This is expected for good quality data. Cells far from the band (for example very high counts with few genes) would be suspicious.

Filtering output. After filtering, the object has 13,714 genes and 2,638 cells. Therefore 62 of the original 2,700 cells (about 2.3%) were removed, so most cells were of good quality and only a few outliers were discarded.

5. Section 3: Normalisation

NormalizeData with the LogNormalize method divides the counts of each gene in a cell by the total counts of that cell, multiplies by 10,000 and then applies a natural log transformation (log(1 + x)). This removes the effect of different sequencing depths between cells so that expression can be compared across cells, and the log transformation reduces the influence of very highly expressed genes. No plot is produced at this step, but all later expression plots (violin plots, feature plots) use these normalised values.

6. Section 4: Highly variable features

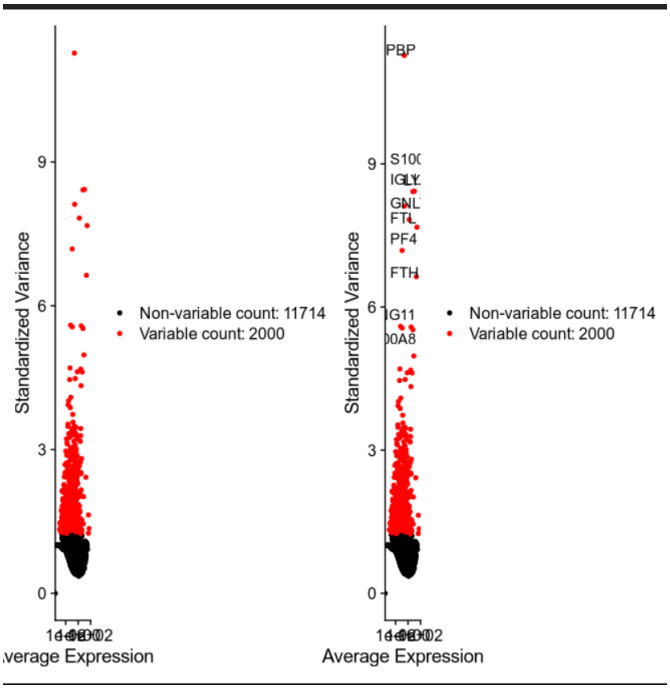


Figure 3: Variable feature plot (left without labels, right with the top 10 genes labelled)

The plot shows average expression on the x-axis (log scale) and standardised variance on the y-axis. Each dot is a gene. The 2,000 red dots are the highly variable genes and the 11,714 black dots are the non-variable genes. Most genes are black and sit near a standardised variance of about 1, meaning their variation between cells is no more than expected for their expression level. The red genes lie above this band, so they differ more between cells than expected, and these are the genes that carry information about cell types.

The labelled genes with the highest variance include PPBP and PF4 (platelet genes), S100A8 (monocyte), GNLY (NK cells), IGLL5 (B cell lineage), and FTL and FTH1 (ferritin genes, highly expressed in monocytes). These belong to different cell lineages, which shows that the main immune cell types drive the variability. In the figure the x-axis tick labels and some gene labels overlap, which is only a display problem and does not affect the analysis. Using only these 2,000 genes for the next steps reduces noise and computing time.

7. Section 5: Scaling

ScaleData was run on all genes. Each gene is centred to a mean of zero and scaled to a variance of one across cells. This gives all genes equal weight in PCA, so very highly expressed genes do not dominate. No plot is produced at this step. Scaling all genes (and not only the variable ones) is needed for the heatmaps later, although PCA itself used only the variable genes.

8. Section 6: PCA (linear dimensional reduction)

Text output: top genes of PC 1 to PC 5

PC	Positive side	Negative side	Interpretation
PC1	CST3, TYROBP, LST1, AIF1, FTL	MALAT1, LTB, IL32, IL7R, CD2	Monocytes/myeloid cells vs T cells
PC2	NKG7, PRF1, CST7, GZMB, GZMA	CD79A, MS4A1, TCL1A, HLA-DQA1, HLA-DQB1	Cytotoxic NK/T cells vs B cells

PC	Positive side	Negative side	Interpretation
PC3	PPBP, PF4, SDPR, SPARC, GNG11	HLA-DQA1, CD79A, CD79B, HLA-DQB1, HLA-DPB1	Platelets vs B cells
PC4	HLA-DQA1, CD79B, CD79A, MS4A1, HLA-DQB1	VIM, IL7R, S100A6, IL32, S100A8	B cells vs T cells/monocytes
PC5	GZMB, NKG7, S100A8, FGFBP2, GNLY	LTB, IL7R, CKB, VIM, MS4A7	NK cells (and some monocyte genes) vs T cells

Each PC is associated with a real biological difference between cell types and not with noise. This confirms that the variable genes contain the information needed to separate the cell populations.

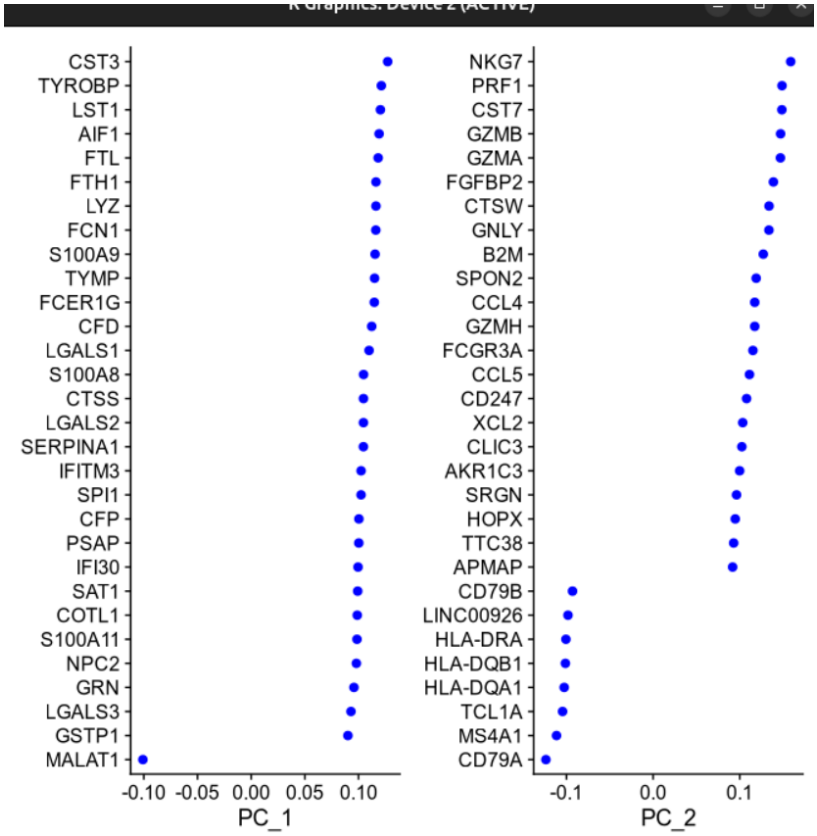


Figure 4: VizDimLoadings for PC1 and PC2

This plot ranks the genes with the largest loadings (contribution) for each PC. For PC1, the positive genes (CST3, TYROBP, LST1, AIF1, FTL, LYZ, FCN1, S100A9) are all myeloid or monocyte genes, while the single strong negative gene is MALAT1, which is relatively higher in lymphocytes. For PC2, the positive genes (NKG7, PRF1, CST7, GZMB, GZMA, GNLY, CCL5) are cytotoxic genes of NK and CD8 T cells, and the negative genes (CD79A, CD79B, MS4A1, TCL1A, HLA-DRA, HLA-DQA1, HLA-DQB1) are B cell and antigen-presentation genes. The loadings of the leading genes are of similar size (about 0.1 to 0.15), so the PCs are driven by groups of co-expressed genes and not by a single gene.

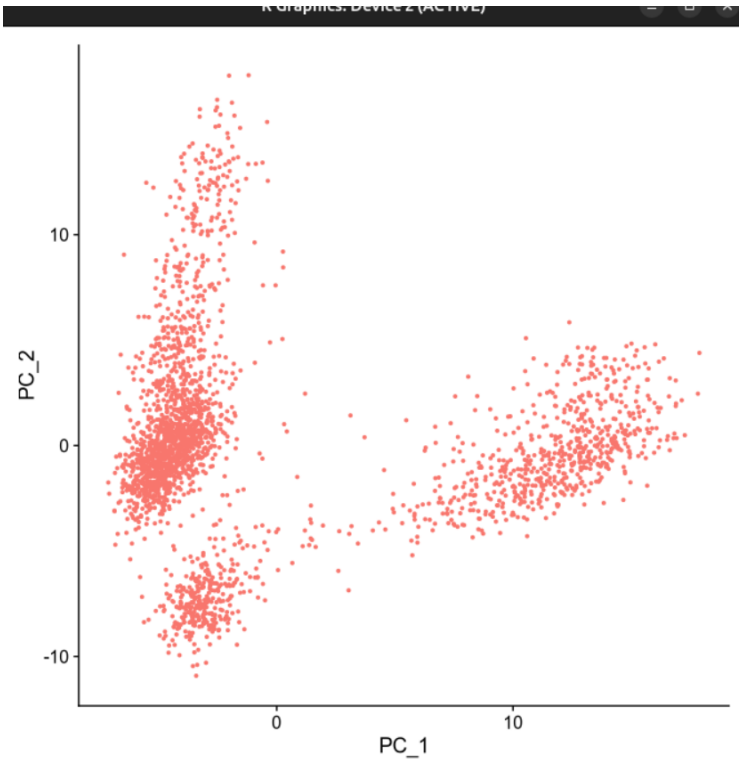


Figure 5: PCA plot (PC1 vs PC2)

The cells separate into three arms. A large dense group lies on the left (negative PC1, with PC2 from about 0 to 15); this is the lymphocyte group (T cells, with the cells towards high PC2 being the cytotoxic NK/CD8 cells). A smaller, tight group at the bottom left (negative PC1, PC2 near -8) corresponds to B cells, which carry the negative PC2 genes. A long spread on the right (positive PC1) corresponds to monocytes and myeloid cells. Even the first two PCs therefore recover the main lineages: T/NK cells, B cells and monocytes. The smaller subtypes are not yet clearly resolved, which is why clustering and UMAP are done next.

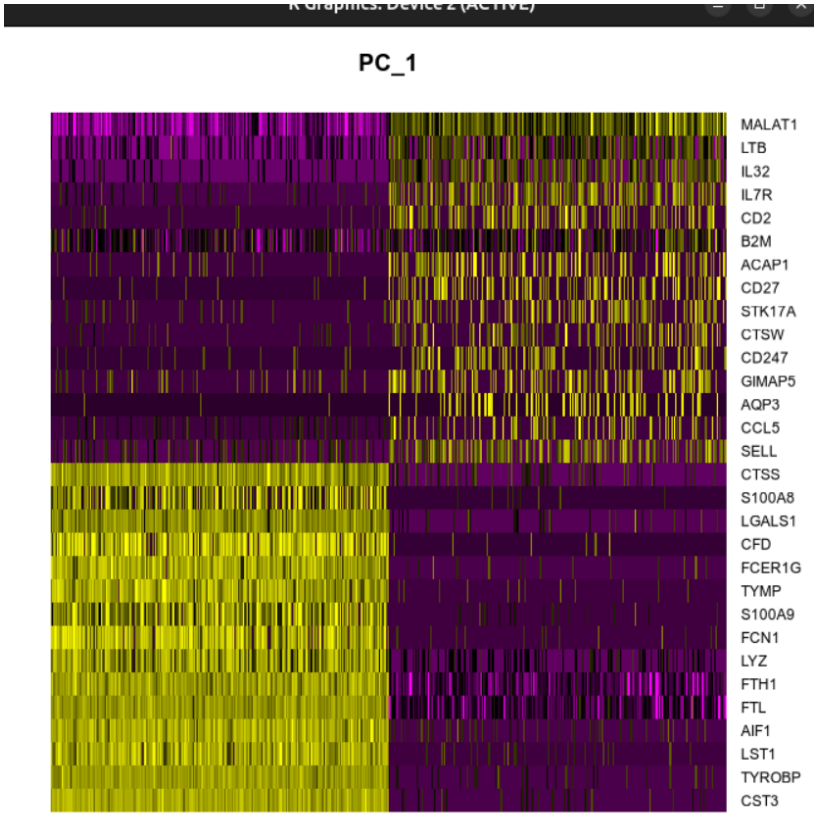


Figure 6: DimHeatmap of PC1 (500 cells at the extremes)

Cells are ordered by their PC1 score along the x-axis and the genes with the strongest loadings are on the y-axis (yellow is high expression, purple is low). The heatmap shows two clean blocks. In the cells on one side, the genes at the bottom (CST3, TYROBP, LST1, AIF1, FTL, LYZ, S100A9, FCN1) are yellow (high) and the genes at the top (MALAT1, LTB, IL32, IL7R, CD2) are purple. In the cells on the other side the pattern is reversed. This confirms that PC1 separates myeloid cells from T lymphocytes and that the genes in each block behave together. The sharp boundary between the two blocks indicates a strong, clear source of variation.

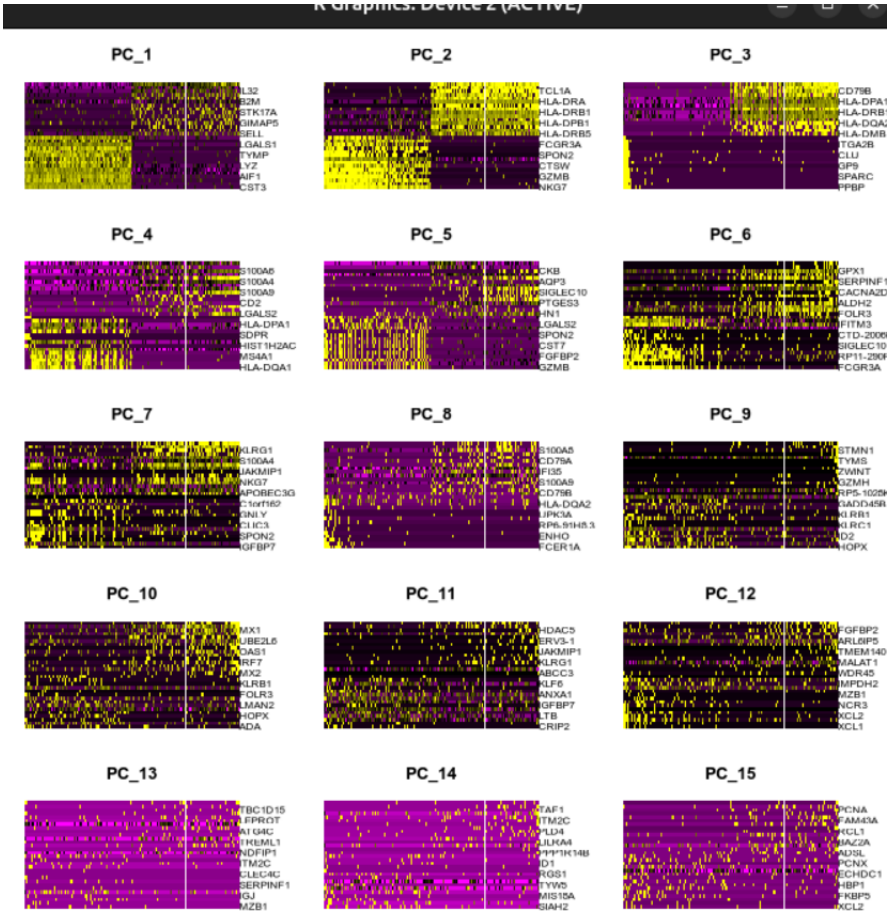


Figure 7: DimHeatmap of PC1 to PC15

The first few PCs (PC1 to about PC5) show clear and distinct blocks of yellow and purple, which means they hold strong structure. PC6 to about PC9 still show some structure but it is weaker and patchier. From about PC10 onwards (especially PC13 to PC15) the heatmaps become noisy and diffuse, with no clear division between the cells at the two ends, so these PCs mostly carry noise or very small effects. This supports using only the first 10 or so PCs.

9. Section 7: Dimensionality of the dataset

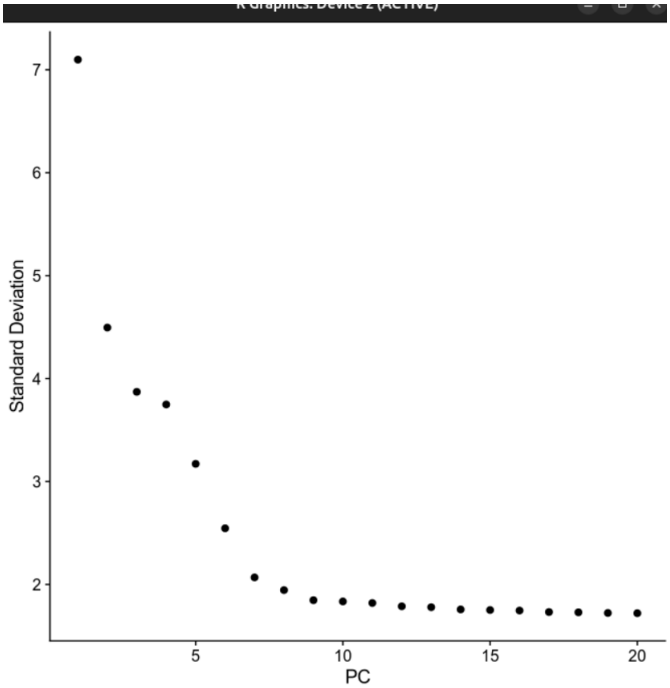


Figure 8: Elbow plot

The elbow plot shows the standard deviation explained by each PC. PC1 has the highest value (about 7.1), followed by PC2 (about 4.5), PC3 (about 3.9), PC4 (about 3.75) and PC5 (about 3.2). The values then drop to about 2.5 at PC6 and about 2.1 at PC7, and after PC9 the curve is flat at about 1.8. The curve therefore bends in the region of PC5 to PC9, and the PCs after about PC9 or PC10 add very little information. Together with the heatmaps in Figure 7, this shows that the first 10 PCs capture almost all the real biological signal, and for this reason 10 PCs (dims = 1:10) were used for clustering and UMAP. Using a few more PCs than the strict elbow is safer than using too few, because it does not discard weaker but real populations such as the small DC and platelet groups.

10. Clustering of the cells

```
Number of nodes: 2638
Number of edges: 95927

Running Louvain algorithm...
0%   10   20   30   40   50   60   70   80   90   100%
[----|----|----|----|----|----|----|----|----|----|
**************************************************|
Maximum modularity in 10 random starts: 0.8728
Number of communities: 9
Elapsed time: 0 seconds
```

Output 4: FindNeighbors and FindClusters (Louvain algorithm)

A nearest-neighbour graph was built with 2,638 nodes (cells) and 95,927 edges (connections between similar cells). The Louvain algorithm then grouped the cells into 9 communities (clusters) at a resolution of 0.5. The maximum modularity in 10 random starts was 0.8728. Modularity measures how strongly cells within a cluster are connected compared with cells outside it; values near 1 indicate very clear community structure, so 0.87 means the clusters are well separated.

```
Maximum modularity in 10 random starts: 0.8728
Number of communities: 9
Elapsed time: 0 seconds
> # Look at cluster IDs of the first 5 cells
head(Idents(pbmc), 5)
AAACATACAACCAC-1 AAACATTGAGCTAC-1 AAACATTGATCAGC-1 AAACCGTGCTTCCG-1
               2                3                2                1
AAACCGTGTATGCG-1
               6
Levels: 0 1 2 3 4 5 6 7 8
>
```

Output 5: cluster IDs of the first five cells

The first five cells were assigned to clusters 2, 3, 2, 1 and 6. The levels are 0 to 8, so there are nine clusters in total. Each cell barcode now carries a cluster label stored in the object.

11. UMAP (non-linear dimensional reduction)

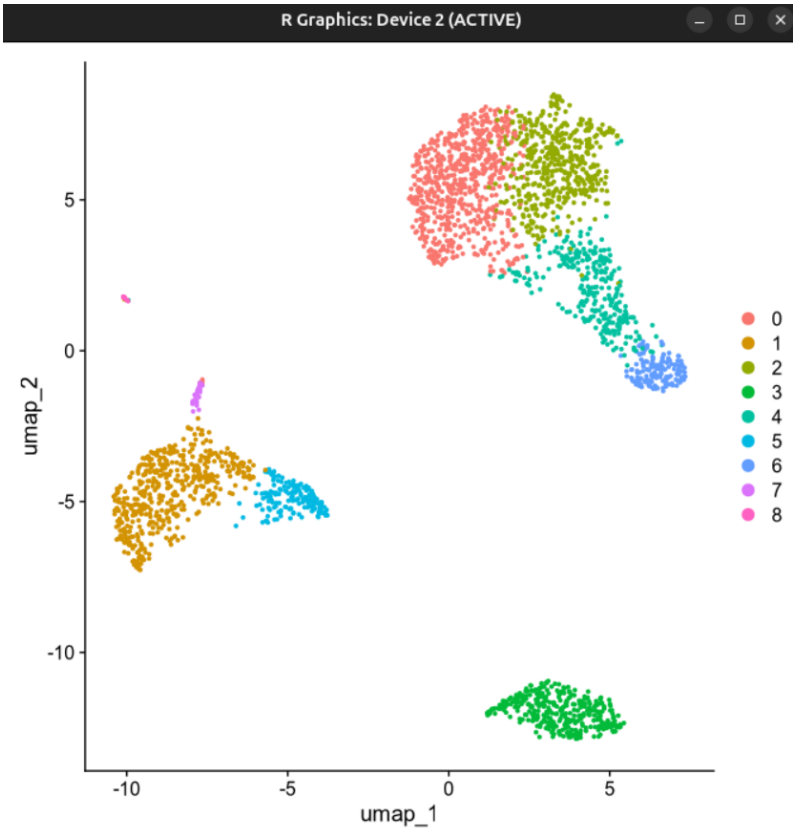


Figure 9: UMAP coloured by cluster (clusters 0 to 8)

UMAP places cells with similar expression profiles close together in two dimensions. The plot shows well separated groups:

- Clusters 0, 2, 4 and 6 form one large connected region in the upper right. Clusters 0 and 2 lie side by side and partly touch, meaning they are very similar cell types. Cluster 4 continues from this region and leads to cluster 6 at the lower right tip, so the cells form a continuum from T cells to cytotoxic cells.
- Clusters 1 and 5 form a second group on the left. Cluster 5 lies directly next to cluster 1, so they are related.

- Cluster 3 forms an isolated island at the bottom, completely separate from the other cells.
- Clusters 7 and 8 are very small. Cluster 7 lies just above the cluster 1 group, and cluster 8 is a tiny isolated group at the far left.

The broad separation into three regions (T/NK cells, monocytes, B cells) agrees with the PCA plot, and the UMAP further splits them into 9 clusters. Distances between separate islands in UMAP should not be interpreted quantitatively; only the grouping and neighbourhood are meaningful.

12. Differentially expressed features (cluster biomarkers)

```
> # find all markers of cluster 2
cluster2.markers <- FindMarkers(pbmc, ident.1 = 2)
head(cluster2.markers, n = 5)
            p_val avg_log2FC pct.1 pct.2    p_val_adj
IL32 2.894376e-90  1.3070772 0.947 0.465 3.969347e-86
LTB  1.061535e-86  1.3312674 0.981 0.643 1.455789e-82
CD3D 8.794641e-71  1.0597620 0.922 0.432 1.206097e-66
IL7R 3.516098e-68  1.4377848 0.750 0.326 4.821977e-64
LDHB 1.642480e-67  0.9911924 0.954 0.614 2.252497e-63
>
```

Output 6: FindMarkers for cluster 2 (top 5 genes)

The top markers of cluster 2 are IL32 (avg_log2FC 1.31), LTB (1.33), CD3D (1.06), IL7R (1.44) and LDHB (0.99), all with extremely small adjusted p-values (about 1e-63 to 1e-86). The columns pct.1 and pct.2 give the fraction of cells expressing the gene inside cluster 2 and in all other cells. For IL7R, 75% of cluster 2 cells express it compared with 32.6% of other cells, and for IL32, 94.7% versus 46.5%. CD3D is a core T cell gene, and IL7R together with LTB and IL32 is characteristic of memory CD4 T cells. Cluster 2 is therefore identified as Memory CD4 T cells. The log fold changes are modest (about 1 to 1.4) because other clusters also contain T cells that share these genes.

```
cluster5.markers <- FindMarkers(pbmc, ident.1 = 5, ident.2 = c(0, 3))
head(cluster5.markers, n = 5)
                    p_val avg_log2FC pct.1 pct.2     p_val_adj
FCGR3A        8.246578e-205   6.794969 0.975 0.040 1.130936e-200
IFITM3        1.677613e-195   6.192558 0.975 0.049 2.300678e-191
CFD           2.401156e-193   6.015172 0.938 0.038 3.292945e-189
CD68          2.900384e-191   5.530330 0.926 0.035 3.977587e-187
RP11-290F20.3 2.513244e-186   6.297999 0.840 0.017 3.446663e-182
>
```

Output 7: FindMarkers for cluster 5 compared with clusters 0 and 3

The top genes are FCGR3A (avg_log2FC 6.79), IFITM3 (6.19), CFD (6.02), CD68 (5.53) and RP11-290F20.3 (6.30), with adjusted p-values of about 1e-182 to 1e-200. These genes are expressed in 84 to 97.5% of cluster 5 cells but in only 1.7 to 4.9% of cells in clusters 0 and 3. FCGR3A (CD16), CD68 and IFITM3 are markers of non-classical (FCGR3A+) monocytes. The fold changes are very large because cluster 5 was compared with clusters 0 (T cells) and 3 (B cells), which are very different cell types; the values would be lower against other monocytes. Cluster 5 is therefore identified as FCGR3A+ monocytes.

```
      p_val avg_log2FC pct.1 pct.2 p_val_adj cluster gene
      <dbl>      <dbl> <dbl> <dbl>     <dbl> <fct>   <chr>
 1 3.74e-112       1.21 0.912 0.592 5.13e-108 0       LDHB
 2 9.57e- 88       2.40 0.447 0.108 1.31e- 83 0       CCR7
 3 1.15e- 76       1.06 0.845 0.406 1.58e- 72 0       CD3D
 4 1.12e- 54       1.04 0.731 0.4   1.54e- 50 0       CD3E
 5 1.35e- 51       2.14 0.342 0.103 1.86e- 47 0       LEF1
 6 1.94e- 47       1.20 0.629 0.359 2.66e- 43 0       NOSIP
 7 2.81e- 44       1.53 0.443 0.185 3.85e- 40 0       PIK3IP1
 8 6.27e- 43       1.99 0.33  0.112 8.60e- 39 0       PRKCQ-AS1
 9 1.16e- 40       2.70 0.2   0.04  1.59e- 36 0       FHIT
10 1.34e- 34       1.96 0.268 0.087 1.84e- 30 0       MAL
# i 7,009 more rows
# i Use `print(n = ...)` to see more rows
>
```

Output 8: FindAllMarkers (positive markers with avg_log2FC > 1, cluster 0 shown)

The printed table lists the top 10 markers of cluster 0 (7,009 more rows exist for the other clusters). They include LDHB (log2FC 1.21), CCR7 (2.40), CD3D (1.06), CD3E (1.04), LEF1 (2.14), NOSIP, PIK3IP1, PRKCQ-AS1, FHIT (2.70) and MAL (1.96), with extremely small adjusted p-values (down to about 1e-108). CD3D and CD3E show that these are T cells, while CCR7, LEF1 and MAL are classical markers of naive T cells, which have not yet met an antigen. Cluster 0 is therefore identified as Naive CD4 T cells. This also explains why clusters 0 and 2 lie side by side in the UMAP: both are CD4 T cells but in different states (naive versus memory).

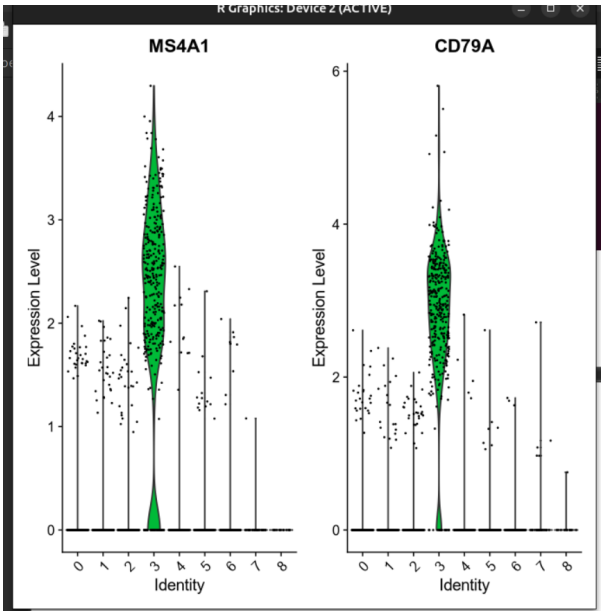


Figure 10: Violin plots of MS4A1 and CD79A (normalised expression by cluster)

Both genes are strongly expressed only in cluster 3: MS4A1 has a median of about 2 to 2.5 and CD79A about 3, while in all other clusters the violins are flat at zero with only a few scattered cells. MS4A1 (CD20) and CD79A (a B cell receptor component) are classical B cell markers, so cluster 3 is identified as B cells. The very specific expression in one cluster shows that these are good, reliable markers.

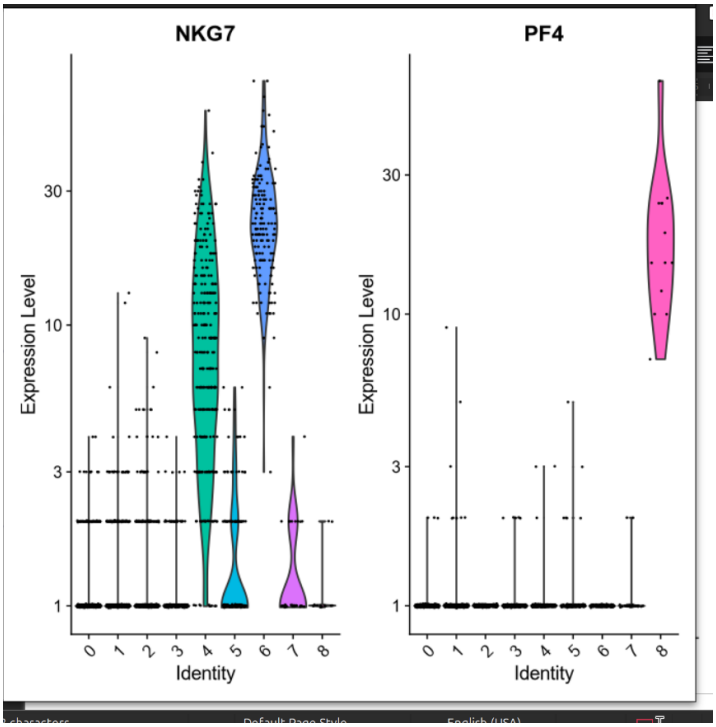


Figure 11: Violin plots of NKG7 and PF4 (raw counts, log scale)

NKG7 is highest in cluster 6 (up to about 40 counts), is also high in cluster 4, and is low or absent in the others. This is typical of cytotoxic cells, with cluster 6 being NK cells and cluster 4 being CD8 T cells. PF4 is expressed only in cluster 8 (counts of about 7 to 45) and is almost absent in all other clusters. PF4 (and PPBP) are platelet-specific genes, so cluster 8 is identified as platelets. The horizontal rows of dots at 1, 2 and 3 appear because raw counts are whole numbers.

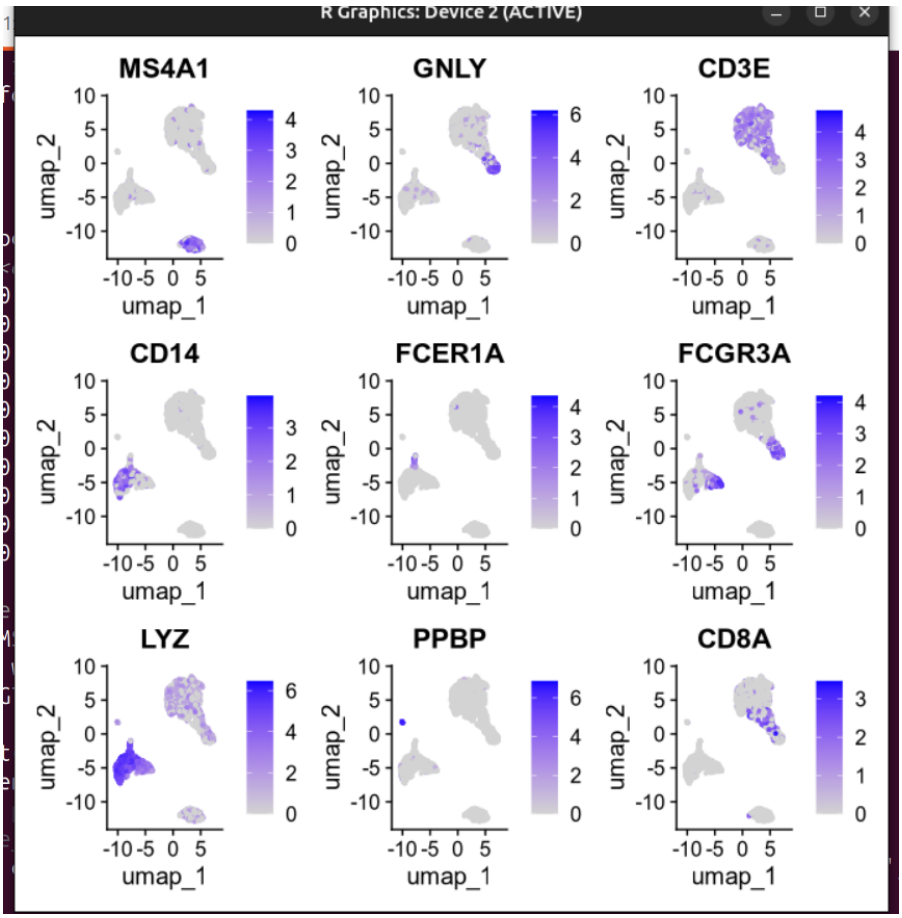


Figure 12: Feature plots of nine marker genes on the UMAP

Each panel colours the cells by the expression of one gene (darker blue means higher expression).

- **MS4A1:** expressed only in the isolated bottom island, the B cell cluster.
- **GNLY:** expressed in the lower right tip of the T/NK region, the NK cluster (6), and slightly in neighbouring cells.
- **CD3E:** expressed throughout the large upper right region, confirming that clusters 0, 2, 4 and 6 are T lineage cells (with lower levels in the NK tip).
- **CD14:** expressed in the left group, mainly the larger part (cluster 1), marking classical monocytes.
- **FCER1A:** expressed only in a very small group near the monocytes, the dendritic cell cluster (7).
- **FCGR3A:** expressed in the smaller part of the monocyte group (cluster 5) and also in the NK tip, since NK cells also express CD16.
- **LYZ:** strongly expressed throughout the monocyte group and also in the small DC group.
- **PPBP:** expressed in a single tiny group, the platelet cluster (8).
- **CD8A:** expressed in the lower part of the T cell region (cluster 4) and partly in the NK tip, in agreement with CD8 T cells.

Each marker gene is restricted to the expected cluster, so the cell-type identification is supported by the expression data and not only by the clustering.

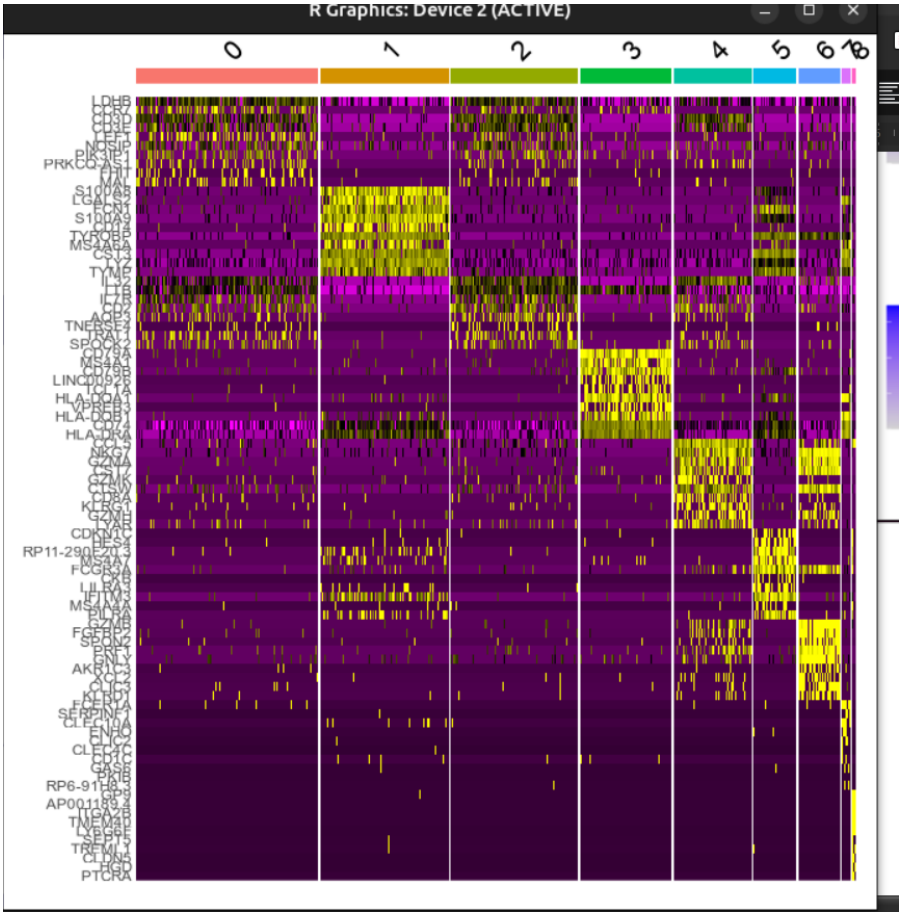


Figure 13: Heatmap of the top 10 markers of each cluster (DoHeatmap)

The columns are cells grouped by cluster (coloured bar at the top) and the rows are the top marker genes. A clear yellow block appears along the diagonal for most clusters, meaning each cluster has its own set of genes that are strongly expressed in it and low in the others.

- Clusters 0 and 2 share T cell genes (CD3D, IL7R, LTB, LDHB), and cluster 0 shows CCR7, LEF1 and MAL. The difference between these two clusters is smaller than between the others, which fits their similar identities.
- Cluster 1 shows S100A8, S100A9, LGALS1, FCN1, TYROBP, CST3 and LYZ (classical monocytes).
- Cluster 3 shows CD79A, MS4A1, HLA-DQA1, HLA-DQB1 and HLA-DRA (B cells).
- Clusters 4 and 6 share cytotoxic genes (NKG7, GZMA, CST7, CTSW); cluster 6 additionally shows GZMB, FGFBP2, PRF1 and GNLY (NK cells).
- Cluster 5 shows FCGR3A, MS4A7, IFITM3 and related genes (FCGR3A+ monocytes), together with some monocyte genes shared with cluster 1.
- Clusters 7 and 8 are very narrow because they contain few cells; their specific genes (DC genes such as FCER1A and CLEC10A for cluster 7, PPBP and PF4 for cluster 8) appear at the bottom as small bright blocks.

The heatmap confirms that the clusters are real biological groups with distinct expression programmes.

13. Assigning cell-type identity to clusters

Using the marker genes found above, the clusters were renamed as follows.

Cluster	Cell type	Key supporting markers
0	Naive CD4 T	CCR7, LEF1, MAL, CD3D, CD3E
1	CD14+ Mono	CD14, LYZ, S100A8, S100A9, FCN1
2	Memory CD4 T	IL7R, IL32, LTB, CD3D
3	B	MS4A1, CD79A, HLA-DQA1
4	CD8 T	CD8A, NKG7, CD3E
5	FCGR3A+ Mono	FCGR3A, CD68, IFITM3
6	NK	GNLY, NKG7, GZMB, PRF1
7	DC	FCER1A, CLEC10A
8	Platelet	PPBP, PF4

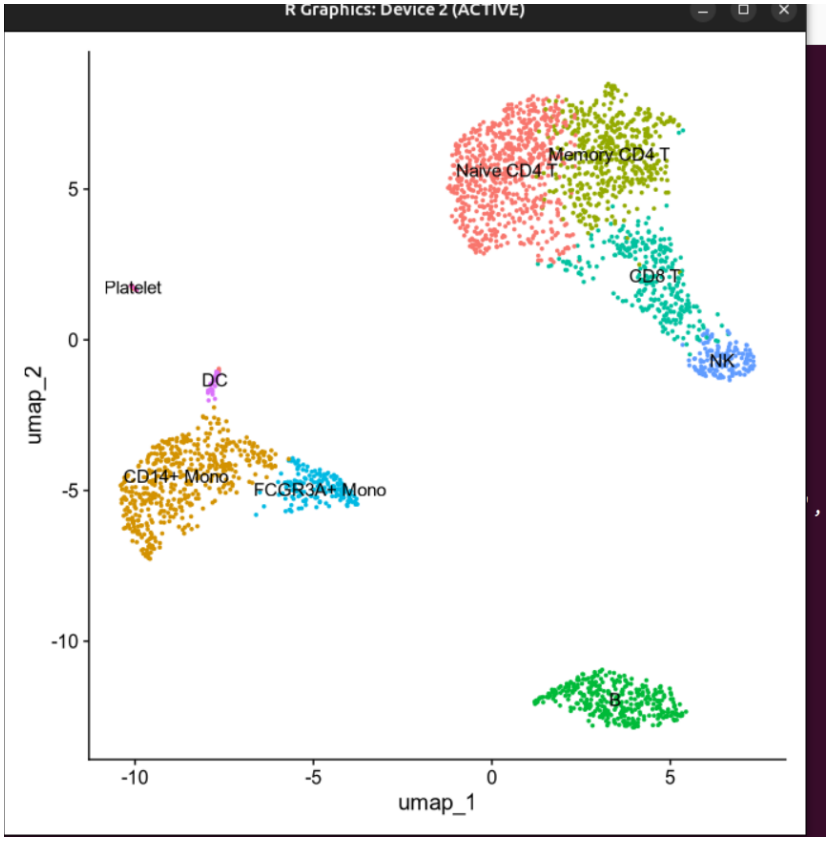


Figure 14: Final UMAP with cell-type labels

The final plot shows the same UMAP as Figure 9, but each cluster now carries its cell-type name. The T cell region (upper right) contains Naive CD4 T, Memory CD4 T, CD8 T and NK cells in a connected arrangement, which is expected because they are closely related lymphocytes. The myeloid cells (CD14+ Mono and FCGR3A+ Mono) form the left group, with the small DC cluster next to them. B cells form a separate island at the bottom, and platelets form a tiny isolated group on the left. The cluster sizes also look reasonable for blood: T cells are the most numerous, followed by monocytes, then B cells and NK cells, while DCs and platelets are rare.

14. Overall conclusion

Starting from 2,700 cells and 13,714 genes, quality control retained 2,638 good quality cells. Normalisation, selection of 2,000 variable genes and PCA captured the main variation, and the elbow plot and PC heatmaps showed that about the first 10 PCs contain the real signal. Louvain

clustering (modularity 0.87) gave 9 clusters, and the marker genes, violin plots, feature plots and heatmap identified them as Naive CD4 T cells, CD14+ monocytes, Memory CD4 T cells, B cells, CD8 T cells, FCGR3A+ monocytes, NK cells, dendritic cells and platelets. These are the expected cell populations of human PBMC, which shows that the workflow was carried out correctly and that the dataset can be resolved into its major immune cell types.